National University of Computer and Emerging Sciences

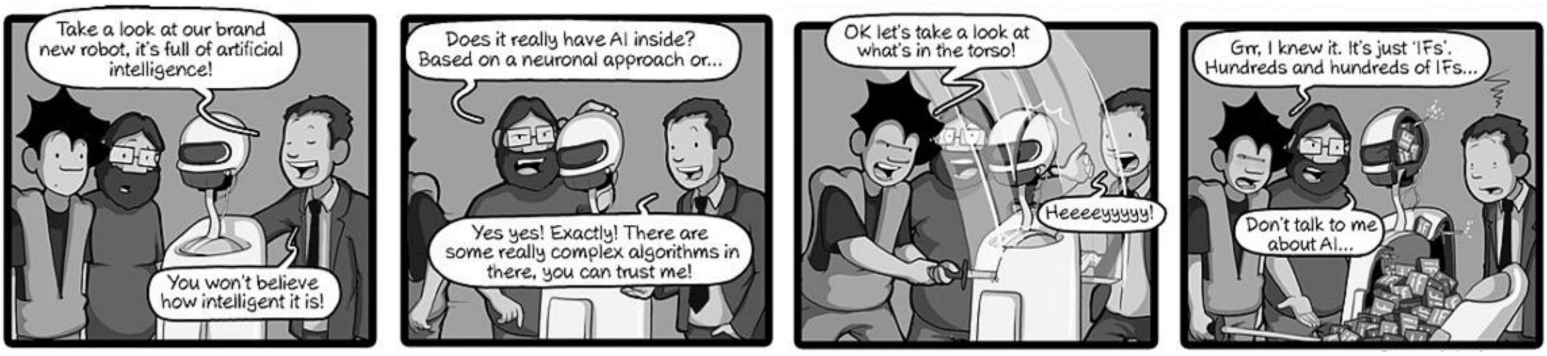
Karachi Campus

Artificial Intelligence (AI2002) Sessional 1 Exam

Date: 26/5/2024 **Total Questions: 02** **Total Marks:** 30

[Marks 12, 3 each, 20 min]

Q1. Consider the comic below and answer the questions that follow: -



A. Which dimension of AI definitions best matches the context in the above comic? Give strong reasons.

MODEL ANSWER:

Thinking Rationally: 1. No mention of action consequences or achieving goal or behaviour. 2. Only mention of complex algorithms but no mention of thinking like a human.

RUBRICS:

1 mark : getting the dimension right

2 marks: for valid reasons.

NOTE: There can be other reasons to justify the answer. If they are correct and support the choice the student has made he can be awarded marks accordingly.

B. What type of Agent did the robot turn out to be? Why were the guys disappointed to find many IFs? Isn't it AI?.

MODEL ANSWER:

It is a simple reflex agent. Although it falls under AI it has limited intelligence and ignores all previous percepts and needs a fully observable.

RUBRICS:

1.5 marks for mentioning simple reflex agent

1.5 marks for the rest of the answer.

C. Give PEAS for solving a jigsaw puzzle.

MODEL ANSWER:

Performance Measure: Correctly assembling the puzzle pieces to form the complete image. Minimizing the time taken to solve the puzzle. Ensuring accuracy (no misplaced pieces). Efficiently exploring possible placements of pieces.

Environment: The puzzle board where pieces need to be placed. The set of jigsaw pieces with unique shapes and colors. A digital or physical workspace where the AI operates. Constraints such as the total number of pieces and edge pieces.

Actuators: If digital: Algorithms that move, rotate, and place pieces in a virtual environment. If physical: Robotic arms or mechanisms that pick and place puzzle pieces.

Sensors: If digital: Image recognition and pattern-matching algorithms to analyze pieces. If physical: Cameras, vision systems, or touch sensors to identify piece shapes, colors, and edges.

RUBRICS:

1 mark for getting the PEAS full form correct.

0.5 marks for each of P E A S.

NOTE: Students only need to either give PEAS for either Physical Version or Digital Version.

D. Which of the categories of Nature of environments does solving jigsaw puzzles belong to? Cover all the characteristics of the task environment. (**HINT**: fully vs partially observable, deterministic vs nondeterministic etc.)

MODEL ANSWER:

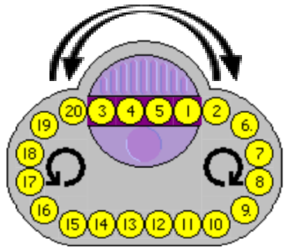
Fully Observable, Deterministic (if digital version else can also be Non-deterministic), Sequential, Static, Discrete, Single Agent (Benign)

RUBRICS:

0.5 marks for each

[Marks 18, 3 each, 40 min]

Q2. The Oval Track puzzle consists of 20 numbered round pieces in one long looped track. Users can slide all the pieces around the loop or use the turntable in the loop which can rotate any four adjacent pieces so that they will be in reverse order. The goal is to arrange the pieces in numerical order from some jumbled-up sequence.



Consider the Oval track puzzle with only 10 numbered round pieces. The tree shows the initial state and next actions with their corresponding states (up to level 1). The state is represented by a string of numbers starting with 4 numbers on the turntable. Imagine the last number is circling back to the first number. E.g. in the initial state the number 10 is adjacent to the number 3.

3 4 5 1 2 6 7 8 9 10

Move turntable   clockwise rotation   anticlockwise rotation

1 5 4 3 2 6 7 8 9 10   **10 3 4 5** 1 2 6 7 8 9   **4 5 1 2** 6 7 8 9 10 3

A. **i.** Complete the search tree up to level 4.

MODEL ANSWER:

L 0		tt			**3 4 5 1** 2 6 7 8 9 10		cw			acw		
L 1	(h = 4)	**1 5 4 3** 2 6 7 8 9 10			(h=5)	**10 3 4 5** 1 2 6 7 8 9			**4 5 1 2** 6 7 8 9 10 3 (h=5)			
	(h=4)	cw	acw	(h=4)	(h=6)	tt	cw	(h=5)	(h=5)	tt	acw	(h=5)
L 2	**10 1 5 4** 3 2 6 7 8 9		**5 4 3 2** 6 7 8 9 10 1		**5 4 3 10** 1 2 6 7 8 9		**9 10 3 4** 5 1 2 6 7 8		**2 1 5 4** 6 7 8 9 10 3		**5 1 2 6** 7 8 9 10 3 4	
	tt	cw	tt	acw	cw	acw	tt	cw	cw	acw	tt	acw
L 3	**4 5 1 10** 3 2 6 7 8 9 (h=6)	**9 10 1 5** 4 3 2 6 7 8 (h=4)	**2 3 4 5** 6 7 8 9 10 1 (h=0)	**4 3 2 6** 7 8 9 10 1 5 (h=4)	**9 5 4 3** 10 1 2 6 7 8 (h=4)	**4 3 10** 1 2 6 7 8 9 5	**4 3 9 10** 5 1 2 6 7 8	**8 9 10** **3** 4 5 1 2 6 7	**3 2 1 5** 4 6 7 8 9 10	**1 5 4 6** 7 8 9 10 3 2	**6 2 1 5** 7 8 9 10 3 4	**1 2 6 7** 8 9 10 3 4 5

RUBRICS:

Since there was a typo in the question it is left with the teacher to judge the approach of the student and give marks.

2 marks for showing that the student understood the problem and showed the relevant actions and resultant states for most nodes.

1 mark for omitting the repeated nodes.

NOTE: If repeated nodes are not removed then the tree is no longer binary and grows very big.

How many nodes of this search tree are visited if using

ii. BFS
MODEL ANSWER: Based on the tree above 12, 26 or 27 is the best answer.
RUBRICS:
Since the state was given wrong (typo) students couldn't have reached the goal state. If they mentioned the total number of states till level 4 (or 3) the answer can be considered correct.
1 mark correct answer (discretion rests with faculty member)
2 mark correct reasoning.
NOTE: Depending upon which action is expanded first from each node the answers can vary but the approach will be the same.

iii. Depth Limited DFS with a limit of 4
MODEL ANSWER: BASED on the tree shown its 7
RUBRICS: 1 mark for correct answer, 2 marks for approach and reasoning
NOTE: Since the limit should have been 3 and NOT 4 in the question. If you find the approach is correct, a full number can be awarded.

B.
i. Suggest a suitable heuristic for the Oval Track Puzzle and show that your heuristic is admissible
MODEL ANSWER: Several Heuristics exists most common are:-

i) misplaced numbers assuming the numbers should start from 1.
ii) count of unaligned numbers.
iii) count of numbers not in increasing order.

These all are not admissible since 1 turntable move can sort 4 numbers thus they all are overestimating.

Utilizing your heuristic how many nodes are visited if using:
ii. A* Search
MODEL ANSWER: Based on the tree above 7 nodes
iii. Greedy first Search
MODEL ANSWER: Based on the tree above 7 nodes
RUBRICS:
3 marks for evaluating correct values for states in the tree.
1.5 marks for A* search
1.5 marks for Greedy search